Polytechnic University of the Philippines
Santa Rosa Campus
Laguna, Philippines

Final Project
FitCheck - A Machine Learning Application Addressing UN SDGs

IT Elective 1
ELEC IT-E1 BSIT 3-1

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February 08, 2025
Date Submitted

I. Intended Learning Outcome:

To design and develop a machine learning-powered application that addresses a societal concern by integrating core concepts such as data preprocessing, model training, and evaluation. This culminates in a functional solution with measurable real-world impact under UN Sustainable Development Goals.

II. Tasks Set-Up, Data and Results and Analysis

Task 1: Problem Identification

Exercise form maintenance is essential for maximizing results and avoiding injuries. Many people work out on their own initiative, depending on personal experience or internet resources, which can occasionally result in improper technique. The squat is one of the most popular exercises, yet it can be difficult to execute properly. To work the right muscle groups and prevent tension on the knees and lower back, proper squat technique is important. Numerous fitness individuals, however, suffer from form-related problems, such as excessive forward tilting, poor back alignment, or insufficient squat depth. People could unintentionally form unhealthy habits that raise their risk of harm and hinder their growth if they don't receive continual feedback.

The interview revealed that the client frequently has trouble evaluating her own squat form. Even after seeing instructions, the client is still uncertain about whether she is performing the exercise correctly. Common problems include trouble balancing and lower back pain brought on by bad posture. The client expressed frustration with the lack of immediate feedback and had a keen interest in a system that offered depth tracking, real-time posture correction, and useful insights.

This gap is intended to be filled by FitCheck, an application driven by machine learning that provides real-time squat form analysis and corrective feedback. Through the application of AI, this tool improves the efficacy and safety of workouts by guaranteeing that users execute squats with the best form possible, lowering the chance of injury, and fostering general physical well-being. FitCheck guarantees an interactive and responsive experience, which makes workouts safer and more efficient than static guidance or pre-recorded workout

videos. From beginners learning correct technique to experienced athletes improving their form, this application has the potential to help a wide range of users.

This program supports injury prevention, safe exercise practices, and improved fitness results, all of which are in line with the Sustainable Development Goal (SDG) 3: Good Health and Well-Being of the UN. FitCheck helps make workouts healthier and more informed by incorporating technology into fitness regimens. This lowers the chance of exercise-related injuries and promotes long-term physical well-being.

Interview: Task 1 interview
Survey results from our target client: FitCheck (Responses)

Task 2: Methodology

- **PROCESS FLOW**

The first flowchart, "Process Flow for Starting viewing the History," symbolizes the general process flow of the FitCheck system. The flow begins as soon as the user opens the application and makes a choice of action: either Start, Restart, Save, or History. If the user selects "Start," they then set a target score and wait for the countdown to start. The system makes sure that the user attains the appropriate starting position before squatting. If the squat depth is incorrect, the user must adjust their form. Progress updates are tracked, and once the target squats are completed, the system displays a congratulatory message.

The second flowchart is "Continuation of Process Flow for Saving the Score," representing the score-saving procedure. If one wants to save, they click the "save" button and pop up a confirmation message. Should it be confirmed, the score gets saved; otherwise, it is canceled, and then it stops and does not save anything. This saves the user's work before closing or quitting the session.

The third flowchart is "Continuation Process Flow for Restarting the Target ," which explains the process of restarting. If the user wants to restart, they should choose to save their score before doing so. A popup confirms this choice and then, by confirming, sets the squat count to zero to let the user set a new target and then start again.

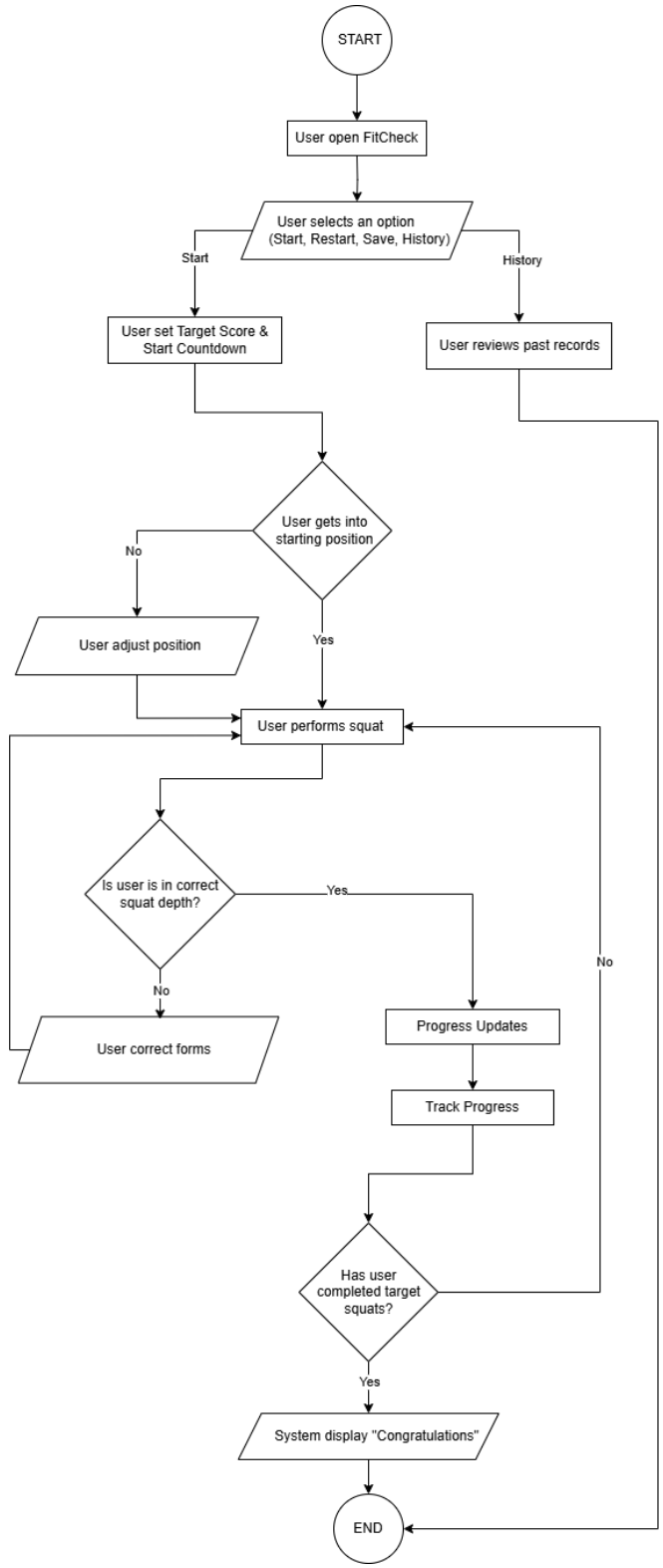

START
User open FitCheck
User selects an option (Start, Restart, Save, History)
Start
History
User set Target Score & Start Countdown
User reviews past records
User gets into starting position
No
Yes
User adjust position
User performs squat
Is user is in correct squat depth?
Yes
No
User correct forms
Progress Updates
Track Progress
Has user completed target squats?
No
Yes
System display "Congratulations"
END

Figure 1. Process Flow for Starting Viewing the History

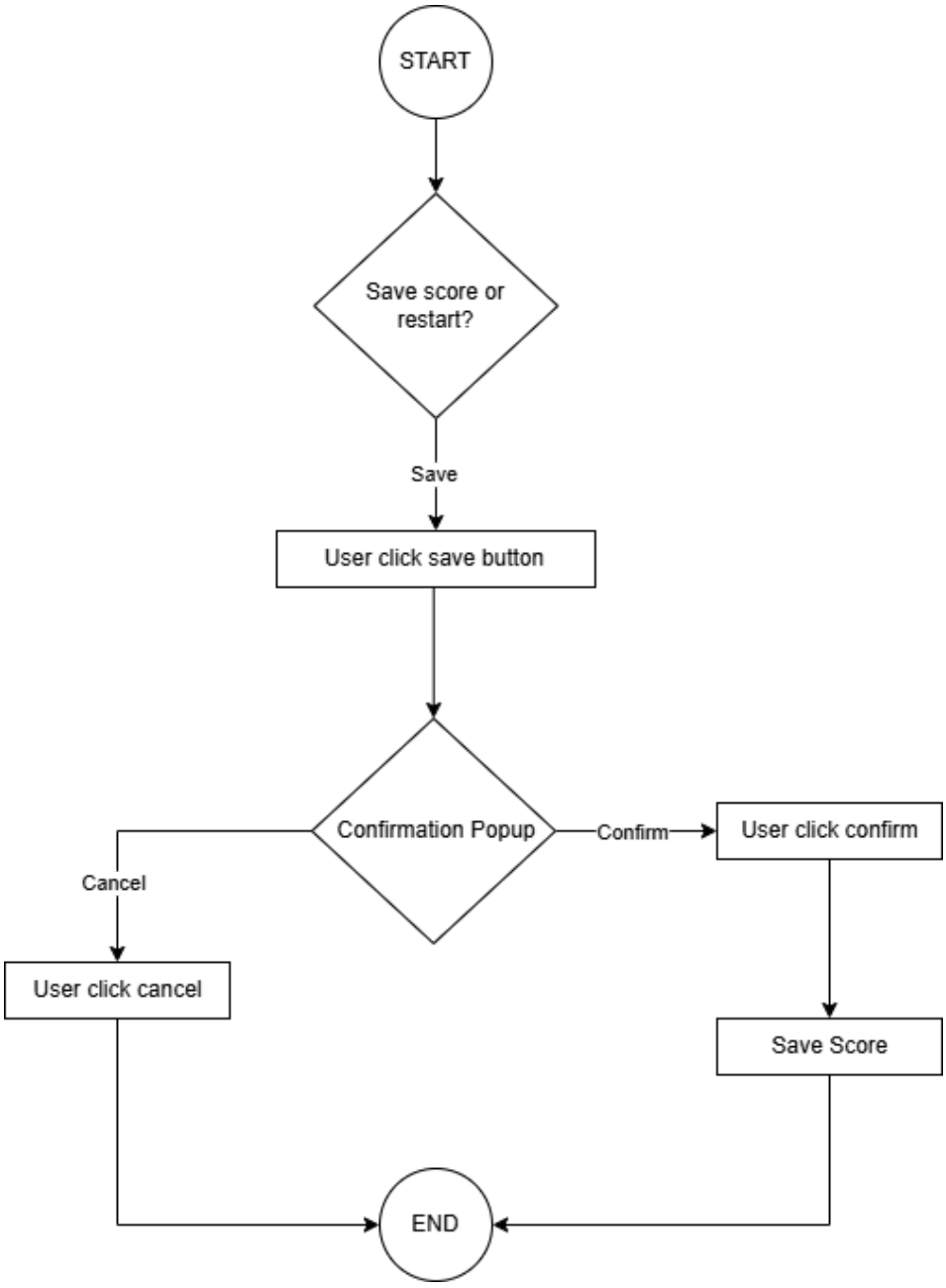


Figure 2. Continuation of Process Flow for Saving the Score

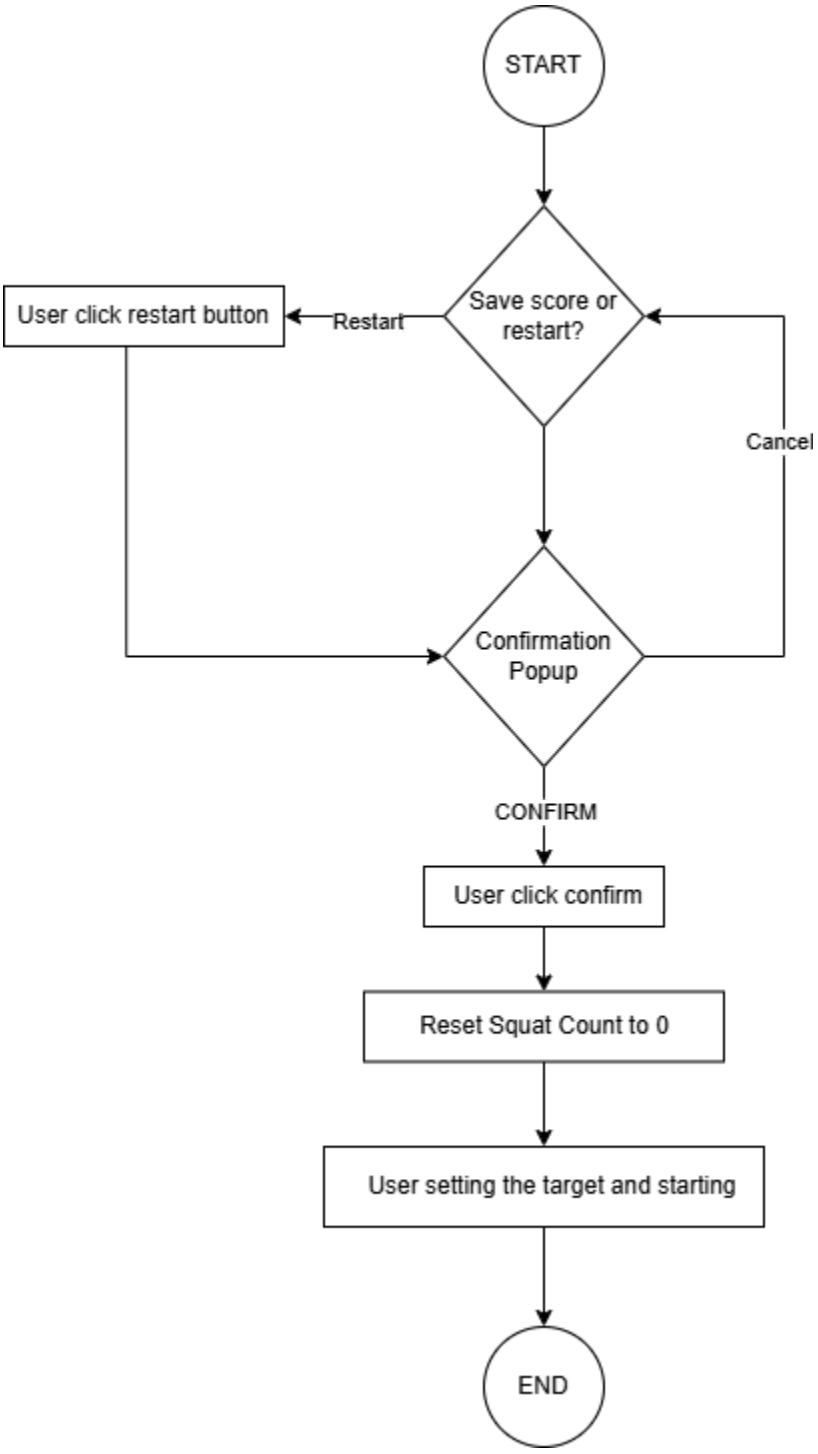


Figure 3. Continuation of Process Flow for Restarting the Targeted Score

- **CODE ALGORITHM**

The FitCheck AI Gym Trainer utilizes a real-time posture detection system combining Python, TensorFlow/Keras, MediaPipe Pose, OpenCV, and Teachable Machine PoseNet to analyze squat form. The system captures video input using OpenCV, processes body landmarks through MediaPipe Pose, and extracts key angles (neck, hip, knee, ankle) to determine squat posture. These extracted parameters are fed into a pre-trained Teachable Machine PoseNet model, which classifies the posture into six categories (c, k, h, r, x, i). The model's predictions are fine-tuned using probability adjustments to improve accuracy. Based on the detected posture, audio feedback is provided using Pygame, where correct squats (c, x) trigger a positive cue, and incorrect postures (k, h, r) prompt a warning. The system tracks the previous posture to avoid miscounting and displays real-time feedback on the video stream with color-coded text and overlays. Additionally, a Flask API enables web-based interaction, allowing users to pause/resume counting, reset the posture count, and mute/unmute audio feedback dynamically. By integrating machine learning with computer vision, the system offers an interactive, hands-free, and accurate squat tracking solution for gym users.

- **SYSTEM FLOW**

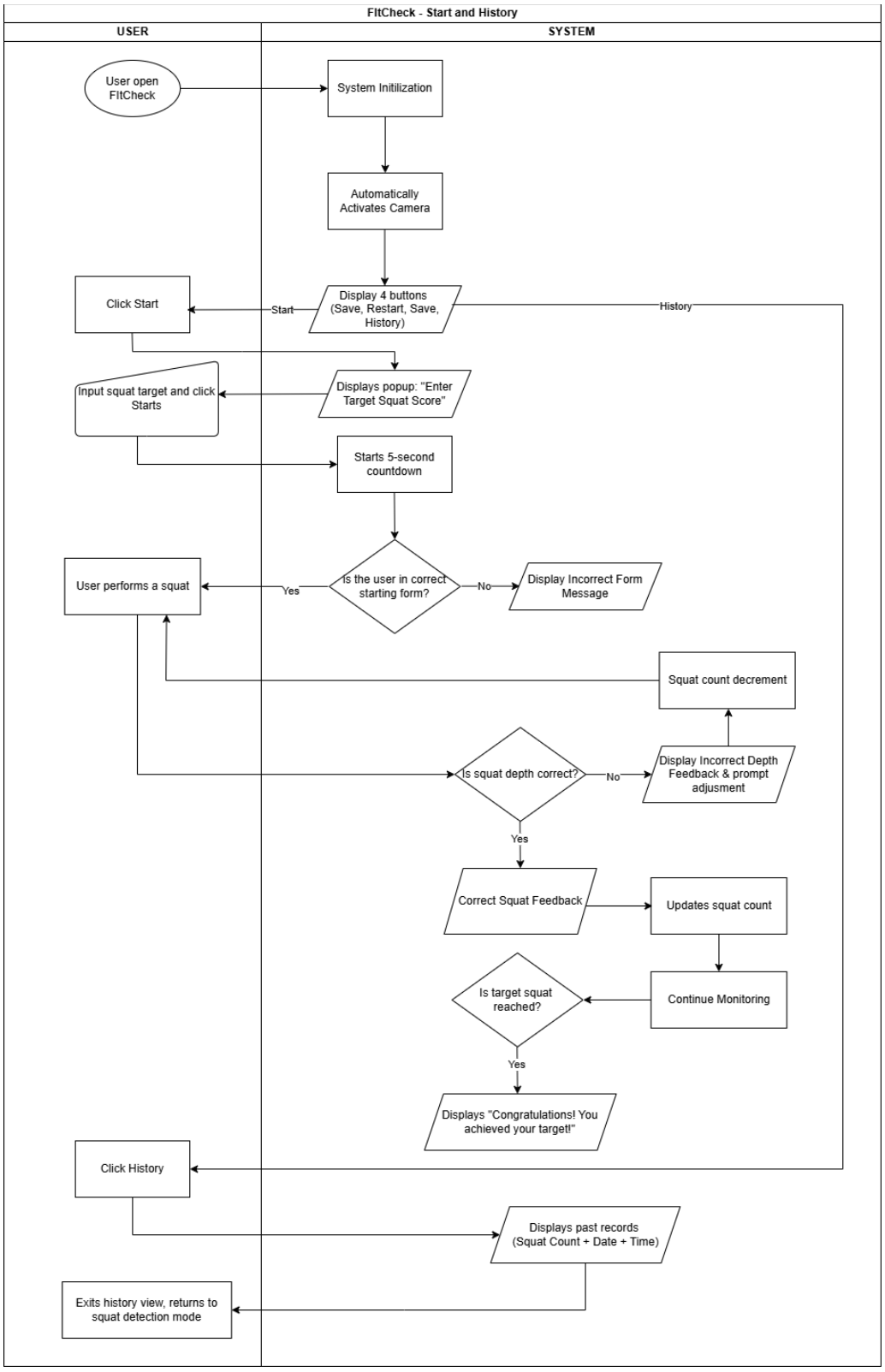

FitCheck - Start and History
USER
SYSTEM
User open FitCheck
System Initilization
Automatically Activates Camera
Click Start
Start
Display 4 buttons (Save, Restart, Save, History)
History
Input squat target and click Starts
Displays popup: "Enter Target Squat Score"
Starts 5-second countdown
User performs a squat
Yes
Is the user in correct starting form?
No
Display Incorrect Form Message
Squat count decrement
Is squat depth correct?
No
Display Incorrect Depth Feedback & prompt adjusment
Yes
Correct Squat Feedback
Updates squat count
Is target squat reached?
Continue Monitoring
Yes
Displays "Congratulations! You achieved your target!"
Click History
Displays past records (Squat Count + Date + Time)
Exits history view, returns to squat detection mode

Figure 4. Start and History View System Flow

The "FitCheck - Start and History" procedure is depicted in the flowchart, which also shows how the user and system interact. The system initializes and turns on the camera automatically when the user opens the FitCheck app. Four buttons are then shown by the system: Save, Restart, Start, and History. After selecting the "Start" button, a popup window requesting the user to enter their desired squat score appears. A 5-second countdown is started by the system after the target has been entered and verified. When the user squats, the system determines whether they are in the proper beginning position. If not, the system analyzes the squat depth; if it does, an improper form alert appears. The system provides positive feedback and updates the squat count if the depth is accurate; if it is incorrect, feedback is given for adjustment. The procedure continues on until the target number of squats is accomplished, at which point the system shows a congratulations message. The system displays historical data, such as the squat count, date, and time, if the user clicks the "History" button. The user can restart the process by returning to squat detection mode after exiting the history view.

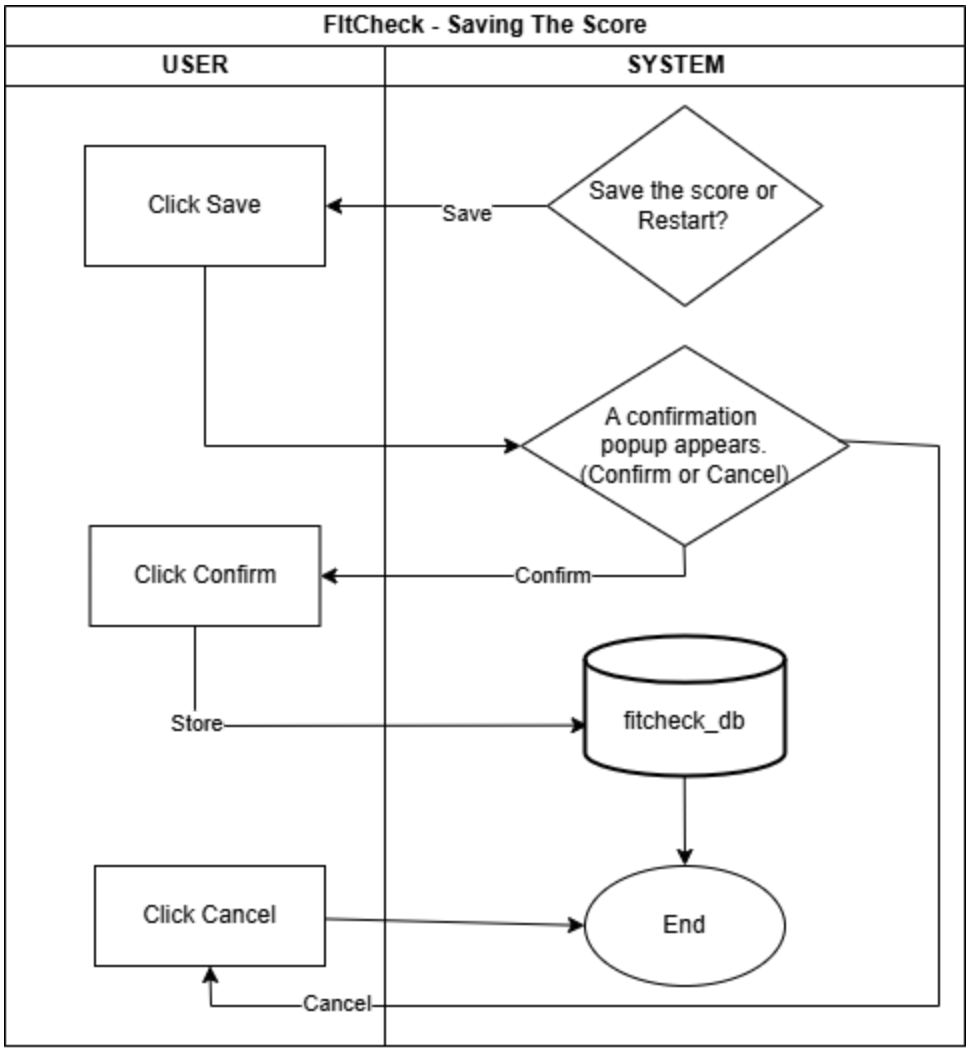


Figure 5. Continuation of System Flow - Saving the Score

The flowchart illustrates the process to save the score in the "FitCheck" system. The user is given the option to either save the score or restart the session by the system. As soon as the user clicks "Save," a confirmation pop up appears and once the user clicks “Confirm” the system saves the accomplished squat count in the "fitcheck_db" database." By doing this, the user's progress is guaranteed to be documented and accessible for review at a later time. As soon as the score is correctly saved, the procedure is over, and the user can utilize the system as needed.

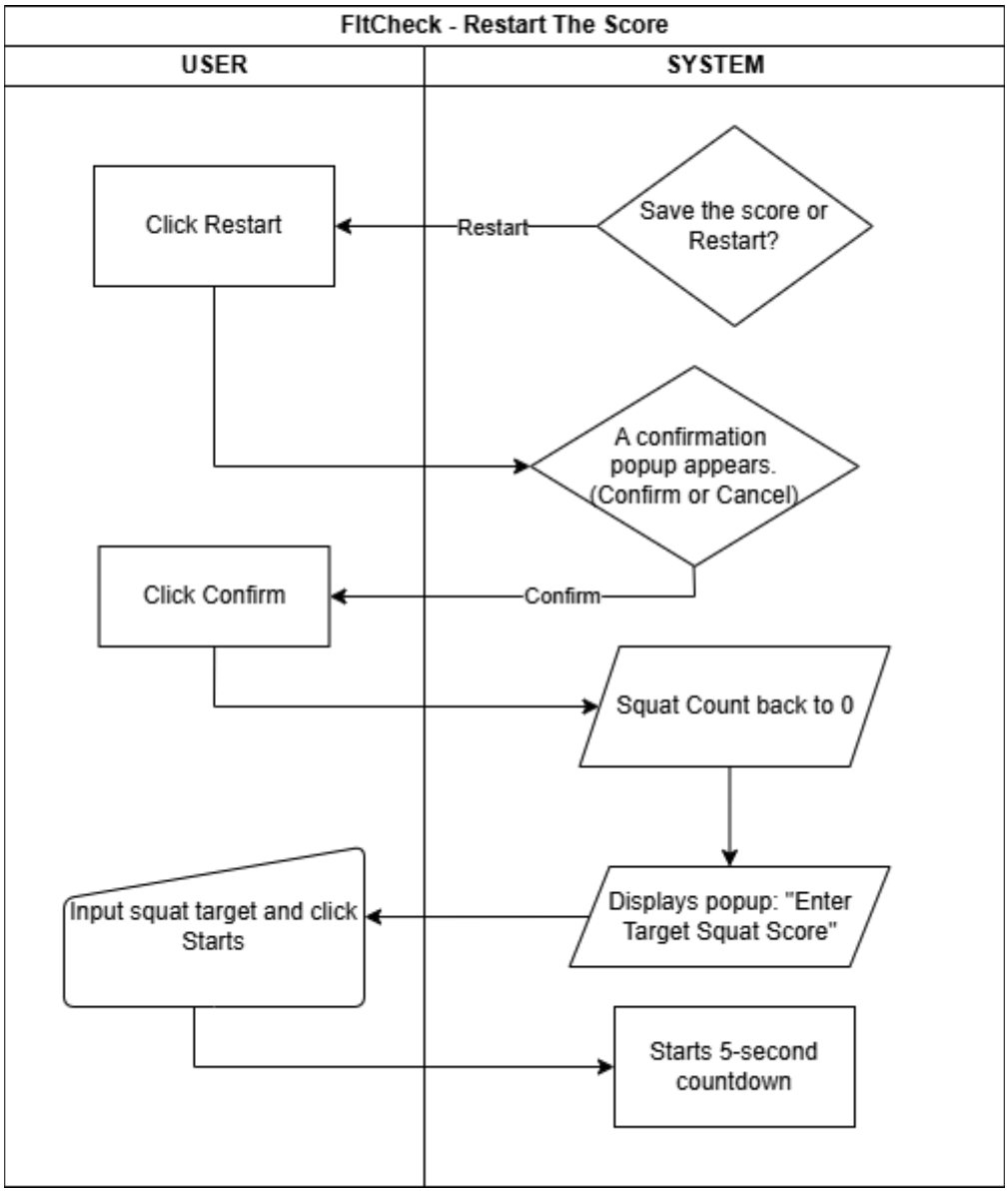


Figure 6. Continuation of System Flow - Restarting the target score

The "FitCheck" system's restart procedure is depicted in the flowchart. The user is given the choice to either save the score or restart the session. If the user chooses "Restart," a confirmation pop up will appear and once the user chooses “Confirm,” the squat count is reset to zero by the system. The user is then prompted to input a new target squat score in a popup window. The system starts a 5-second countdown once the user enters the target and selects "Start," and then starts a fresh squat detection session. By establishing fresh objectives, this procedure enables the user to monitor and enhance their performance over time.

TASK 3: ML Dataset Gathering, Modeling, and Model Performance Test and Evaluation

The dataset used for the posture checker model was collected and processed based on various joint coordinates extracted from squat movements. The dataset includes five key joint positions: neck, knee, hip, ankle, and Y-Knee (Vertical Position of the Knee).

The dataset contains 500 samples each for the classes correct squat, misaligned knees, back rounding (excessive forward lean or lower back rounding), and insufficient depth. The data was collected from Kaggle, a platform for datasets in various fields. Majority of the data were video-based tracking and motion capture methods, where joint positions were extracted using an AI model.

The AI model was trained using TensorFlow/Keras, focusing on classifying squat form based on joint position data. The training parameters is presented in the table below:

Parameter	Value
Epochs	50
Batch size	16
Learning Rate	0.001
Evaluation Metrics	Accuracy, Precision, Recall

Table 1. Training Parameters

In Table 1, An epoch is one full pass of the entire dataset through the model, and the model trains for 50 cycles to refine its ability to classify squat posture correctly. The Batch size is the number of samples processed before updating the model weights. Its value is 26 to balance memory efficiency and model convergence speed. The learning rate controls how much the model updates weights after each step. Its value is 0.001 to prevent drastic changes in weights, leading to stable training. Evaluation metrics are essential for assessing model performance, guiding improvements, and ensuring optimal prediction accuracy. They help measure reliability, identify relevant patterns, and support decisions on model selection and tuning.

The model demonstrated dependable squat classification with an accuracy of 89.5%. The model successfully distinguishes between proper and improper squat posture, as indicated by precision and recall scores that indicate acceptable classification balance. There are certain misclassifications, most likely because of borderline postures, in which the squat form is almost accurate but not quite.

Task 4: Implementation and Development

This documentation provides a concise overview of the development and implementation of a machine learning-powered application designed to address issues concerning improper squatting. The app processes inputs, applies a trained ML model, and delivers meaningful insights to users.

Test cases with the client and video recordings were used to document the app's functionality. By verifying that the predictions, analysis, and outcomes were presented accurately, these tests verified the system's functionality. The app's dependability was increased by fixing problems that arose during testing.

App functionality:

FitCheck - AI Gym Trainer - School - Microsoft Edge 2025-02-08 10-26-50.mp4

Demo: demo

Test cases with the client:

The test case for the FitCheck AI Gym Trainer System was conducted to validate the accuracy and reliability of its real-time posture classification and feedback mechanisms. The system was initialized with a connected camera, and the Flask application was launched to serve the video feed. The test case required users to perform various squat postures, including correct form, knee misalignment, improper back positioning, and insufficient depth, while the system processed the movements using Mediapipe for pose detection and a pre-trained TensorFlow model *(working_model_1)* for classification. The system's response was evaluated based on the displayed posture classification, incremental updates to the improper count, and audio feedback provided for each detected posture. To ensure consistency, the */improper_count* API endpoint was tested for real-time tracking accuracy. The system was deemed successful as it consistently classified postures correctly, provided immediate auditory cues, and maintained an accurate count of improper postures. However, failure conditions, such as misclassifications, improper count errors, or system crashes, were monitored to assess limitations.

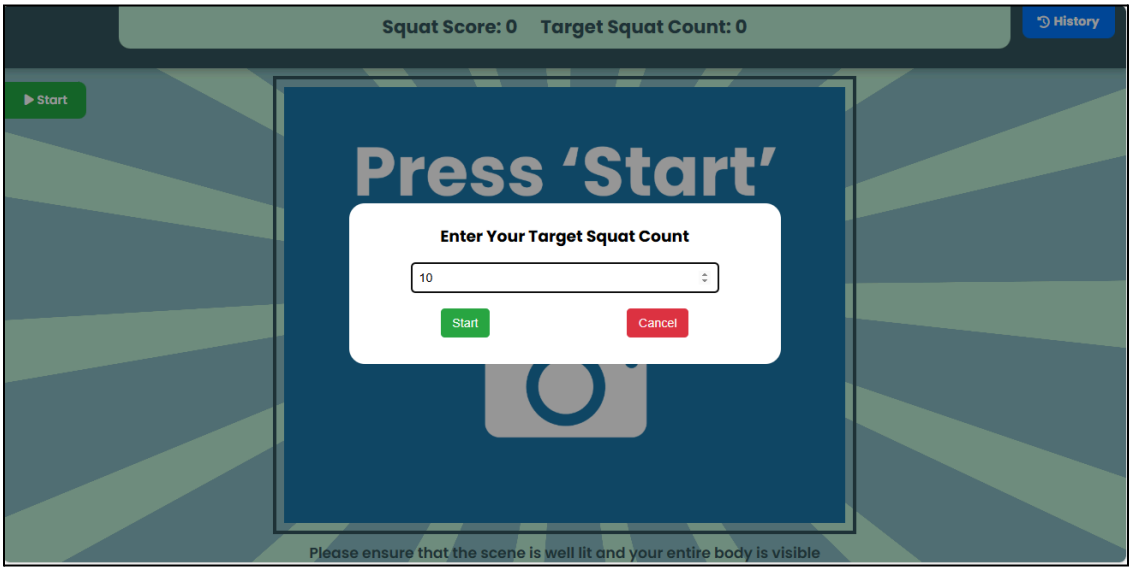
Real-time body motions are recorded by MediaPipe, which also normalizes the data and identifies important landmarks. TensorFlow-Keras determines posture as either correct or incorrect based on the processed data. TensorFlow-powered Keras makes model inference and training easier. The predictions are then utilized to give users visual feedback in real time, enabling them to rapidly correct their posture. FitCheck posture correction is guaranteed to be precise, effective, and scalable with the help of its smooth integration.

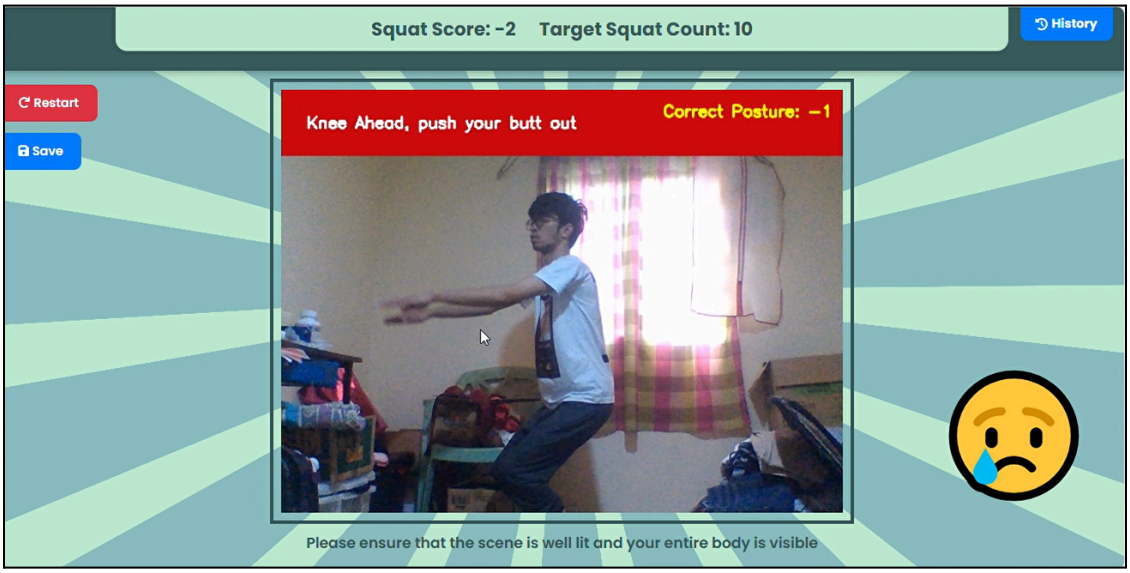
Test Case	Expected Outcome	Actual Outcome	Pass/Fail
User performs a correct squat	System classifies as "Correct Form"	Correctly classified	Pass
User moves knees forward	System detects "Knee Ahead" error	Correctly classified	Pass
User rounds back incorrectly	System detects "Back Wrongly Positioned" error	Correctly classified	Pass
Audio feedback plays for errors and correct squat	Correct audio plays	Correct audio output for proper and improper squat	Pass
Squat score updates accurately	Count updates on squat score	Accurate tracking	Pass

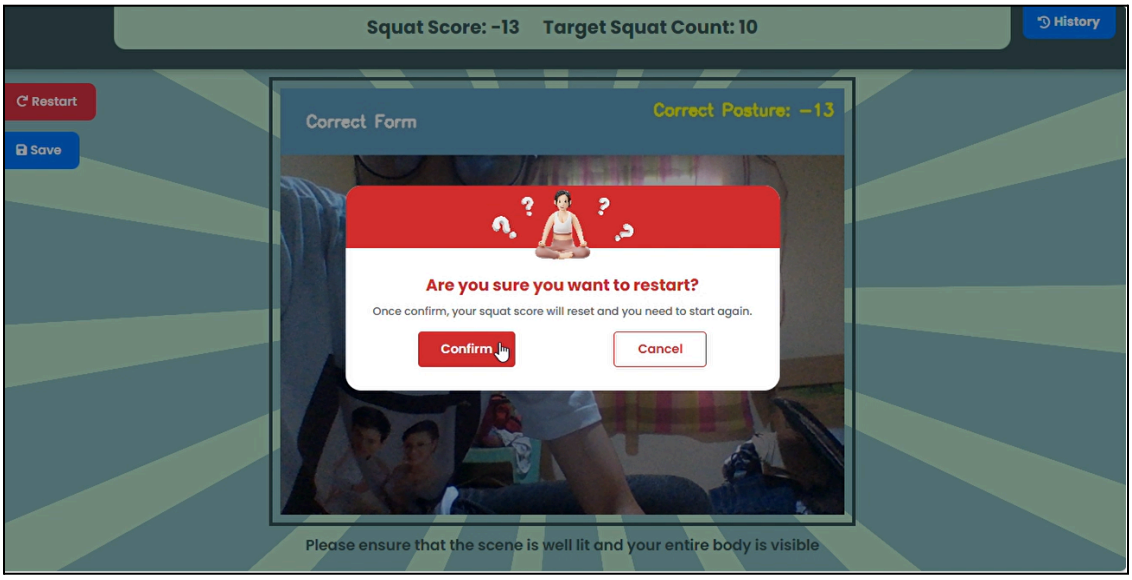
Table 2. Test Cases and Outcomes

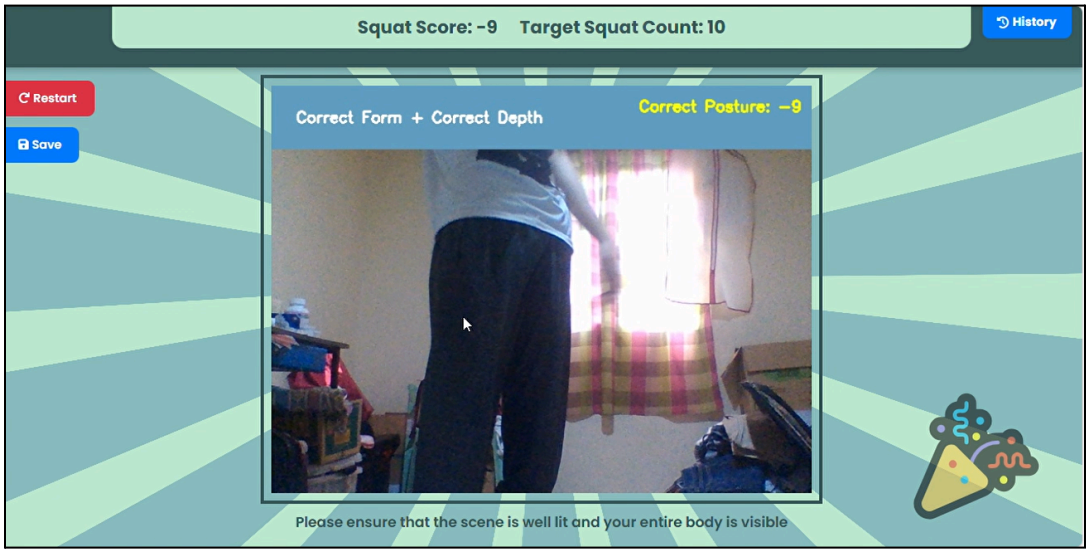
FitCheck User Interface

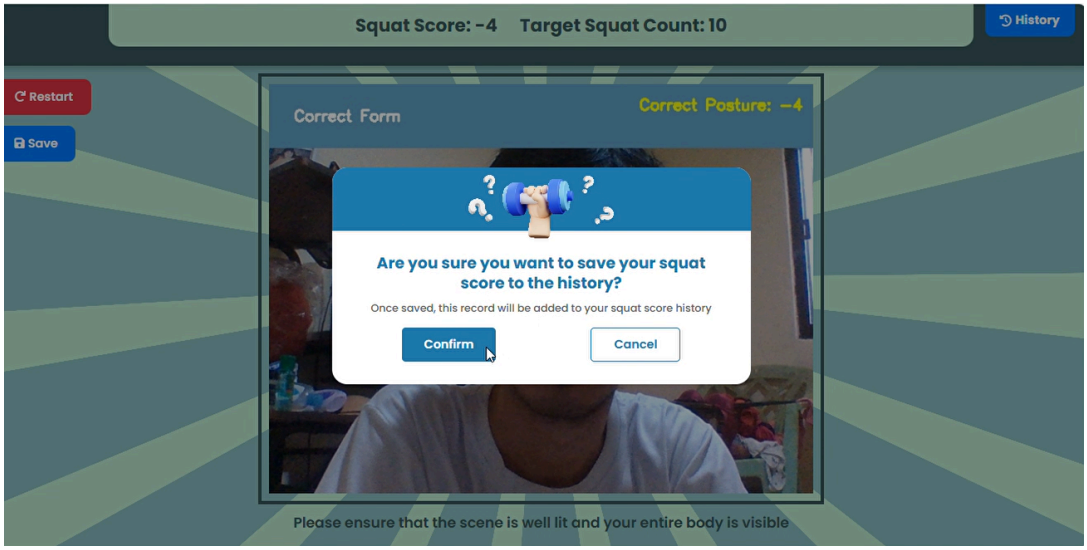


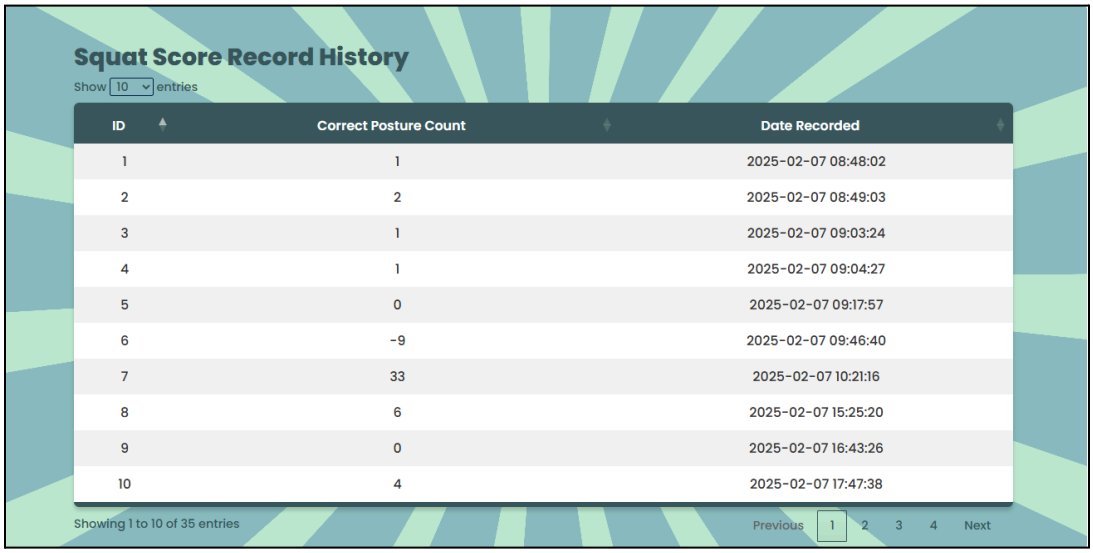


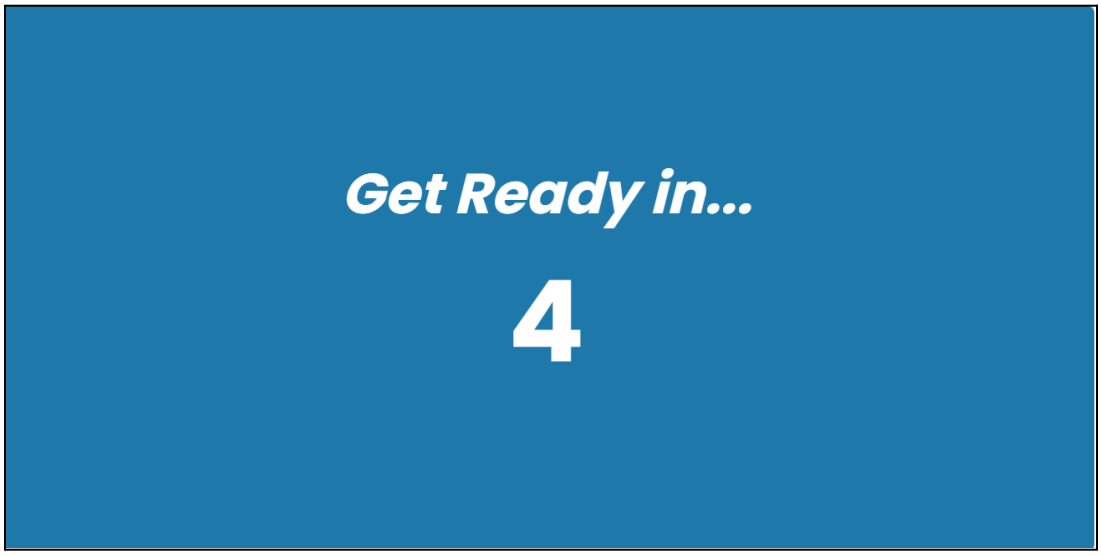

Squat Score: -2 Target Squat Count: 10
History
Restart
Save
Knee Ahead, push your butt out
Correct Posture: -1
Please ensure that the scene is well lit and your entire body is visible

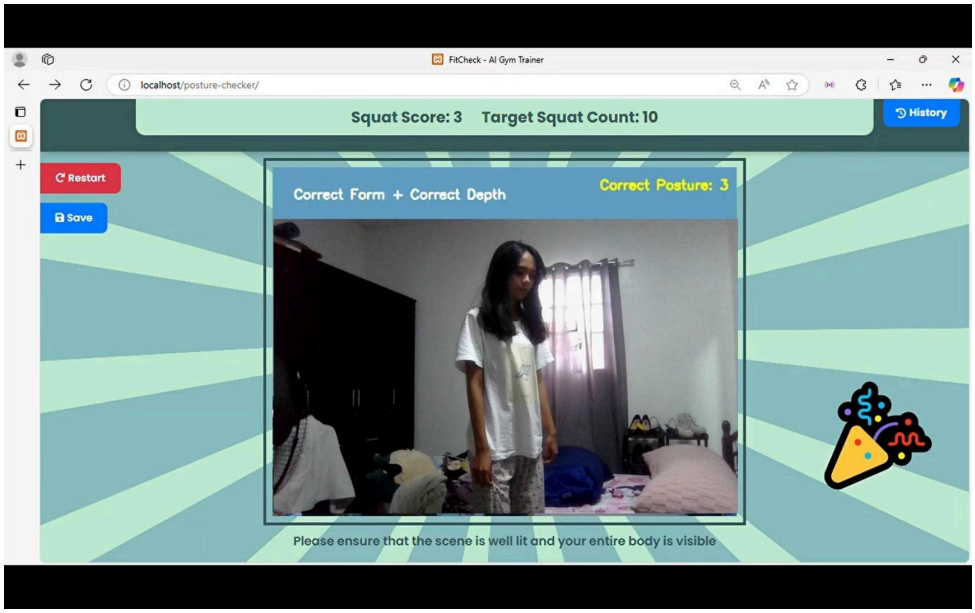

Squat Score: -13 Target Squat Count: 10
History
Restart
Save
Correct Form
Correct Posture: -13
Are you sure you want to restart?
Once confirm, your squat score will reset and you need to start again.
Confirm
Cancel
Please ensure that the scene is well lit and your entire body is visible

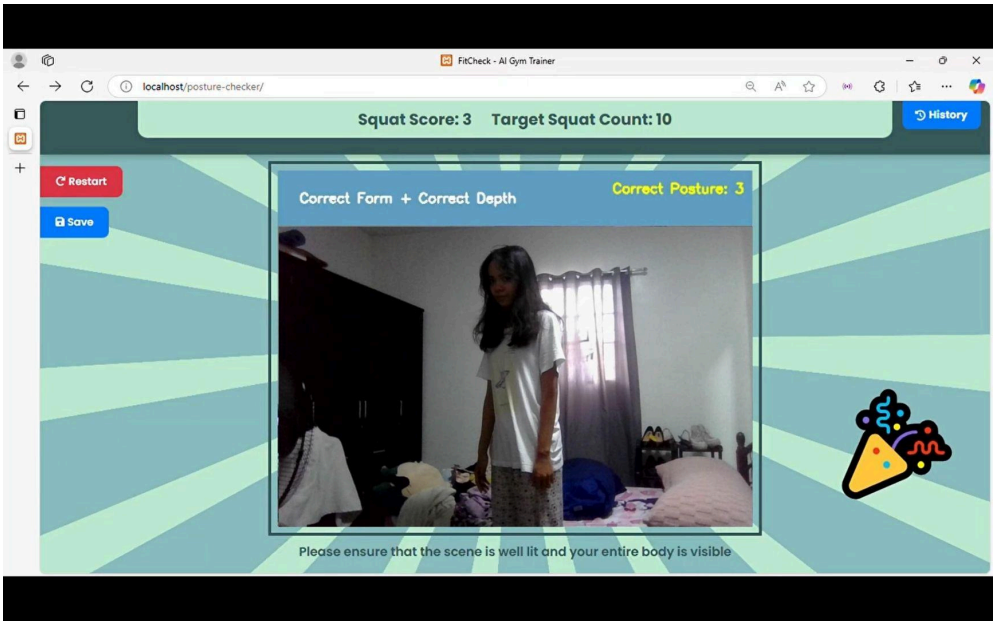

Squat Score: -9 Target Squat Count: 10
History
Restart
Save
Correct Form + Correct Depth
Correct Posture: -9
Please ensure that the scene is well lit and your entire body is visible


Squat Score: -4 Target Squat Count: 10
History
Restart
Save
Correct Form
Correct Posture: -4
Are you sure you want to save your squat score to the history?
Once saved, this record will be added to your squat score history
Confirm
Cancel
Please ensure that the scene is well lit and your entire body is visible


Squat Score Record History
Show 10 entries
ID
Correct Posture Count
Date Recorded
1 1 2025-02-07 08:48:02
2 2 2025-02-07 08:49:03
3 1 2025-02-07 09:03:24
4 1 2025-02-07 09:04:27
5 0 2025-02-07 09:17:57
6 -9 2025-02-07 09:46:40
7 33 2025-02-07 10:21:16
8 6 2025-02-07 15:25:20
9 0 2025-02-07 16:43:26
10 4 2025-02-07 17:47:38
Showing 1 to 10 of 35 entries
Previous 1 2 3 4 Next


Get Ready in...
4


FitCheck - AI Gym Trainer
localhost/posture-checker/
Squat Score: 3
Target Squat Count: 10
History
Restart
Save
Correct Form + Correct Depth
Correct Posture: 3
Please ensure that the scene is well lit and your entire body is visible


FitCheck - AI Gym Trainer
localhost/posture-checker/
Squat Score: 3
Target Squat Count: 10
History
Restart
Save
Correct Form + Correct Depth
Correct Posture: 3
Please ensure that the scene is well lit and your entire body is visible

Task 5: Deployment

This document outlines the deployment of the FitCheck: Squat Posture Correction System and details the client presentation, feedback collection, and areas for improvement. The system's main features and purpose were explained to the client during their introduction. During a live presentation, the system assessed squat posture and gave immediate feedback. The client engaged with the demonstration to assess the system's usability. FitCheck was beneficial to the client, particularly for those who exercise alone and have no access to proper form instructions for squats. The most useful element, which gave the impression of a virtual coach, was the real-time posture adjustment. Some of the client's feedbacks include that the system was easy to grasp, the UI was simple, and errors like leaning forward, improper knee alignment, and problems with squat depth were successfully identified by the algorithm. One significant benefit, especially for beginners, was the real-time corrections.

The client also suggested some features for improvement such as an audio guide corrections in place of users having to continuously glance at the screen, a tracking feature to keep tabs on advancements over time, FitCheck's usability may be improved by enabling integration with workout applications., and extending the program to assess additional activities, such as lunges and deadlifts.

The demonstration of the app's functionality to the client, and the client feedback is embedded within the links below:

- Task 5 - demo with the client.mp4 - Demonstration of the system and the client feedback
- FitCheck (Responses) - survey Results in Sheet format.
- FitCheck Survey

Task 6: Presentation

IEEE Poster

FIT CHECK: A Machine Learning Application Adressing UN SDGs

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Abstract

FitCheck is a machine learning-powered app designed to encourage people to work out correctly, thus reducing the risk of pain and injury while working out at home. The project corresponds to UN Sustainable Development Goal 3: Good Health & Well-being, as musculoskeletal strain throughout exercise, such squatting, remains one of the main concerns in the fitness world. FitCheck uses computer vision techniques to measure body position, assess position, and give the user real-time feedback during safe and effective exercise. The system was trained on a dataset of different squat positions categorized as correct and incorrect forms. The model was tested for reliability by accuracy, precision, and recall metrics. The results show that the system could detect poor form in squatting and provide real-time corrective instruction. This approach can be taken to a fitness monitoring and injury prevention level with a view toward people exercising more safely without an individual trainer. By using machine learning and computer vision, FitCheck is contributing to the global effort of promoting accessible and injury-free physical activity, thereby enhancing overall well-being.

Keywords

FitCheck; machine learning-powered up; musculoskeletal strain; work out correctness; and injury prevention

I. Introduction

[1] In the early 2000s, health monitoring was often limited to basic medical equipment and sporadic doctor visits. The terrain has changed significantly in the modern era due to the arrival of advanced technologies. According to a notable McKinsey & Company report, 76% of consumers are now willing to use digital health solutions as a result of the increase in smartphone penetration, which reached 76% among individuals in 2021. [2] Artificial intelligence and machine learning are becoming an ordinary aspect of human life in the 21st century. Gartner's researchers have classified AI as a top strategic technology trend. [3] Exercise at home has several benefits. You always have access to the equipment you need, it's economical, and it can be a great way to spend quality time with your loved ones. However, training injuries can occur at any time, including at home, particularly if you're lifting weights or performing a high-intensity exercise with poor technique and posture. Additionally, some individuals have a tendency to overwork themselves at home, which can cause muscle strains and/or exacerbate pre-existing ailments like knee or joint discomfort.

To address the challenge, FitCheck is a computer vision application in machine learning technology that uses machine learning algorithms for analyzing squat posture, providing assessments of movement accuracy, and feedback in real-time. Using such a dataset consisting of multiple forms of squat postures and the categorization between correct and incorrect forms, this system is ready to offer detailed corrective instructions that will further assist in increasing exercise safety and efficiency.

This method is in tandem with the United Nations Sustainable Development Goal (SDG) 3: Good Health & Well-being since it encourages safe physical activity without an increased risk of workout-related injuries.This research will evaluate whether FitCheck efficiently and correctly identifies bad squatting posture and provides adjustment suggestions. In this sense, the paper evaluates the performance of the system in terms of precision, recall, and accuracy and which will identify its validity for application in practice. The course of the research is expected to contribute to the further development of AI-based fitness monitoring technologies that make exercise safe, accessible, and injury-free for all people around the world.

II. The Client, Problem and SDG

The main client for this project, Ms. Athena Gracia Naag, is a person who actively participates in home workouts as part of her fitness routine. She particularly performs squats to increase her overall endurance and build her lower body. Her biggest obstacle is keeping up good squat form because bad posture and movement patterns can significantly raise the risk of long-term pain, discomfort, or even major injury.In order to ensure safe and efficient exercise performance, this problem requires a solution that can give real-time feedback and remedial instruction to provide a safe, efficient, and effective solution. In line with these issues, our project directly contributes to SDG 3: Good Health & Well-Being in that our project will prevent injury improve the general physical fitness of users, and promote sustainable exercise habits through embracing state-of-the-art machine learning and computer vision approaches.

III. Methodology

A. Process Flow

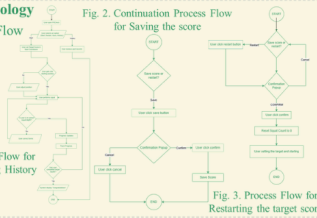


Fig. 1. Process Flow for Starting viewing History

Fig. 2. Continuation Process Flow for Saving the score

Fig. 3. Process Flow for Restarting the target score

B. System Flow



Fig. 4. Start and History System Flow

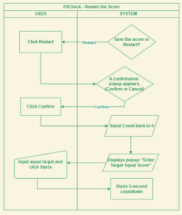


Fig. 5. Restarting the target score

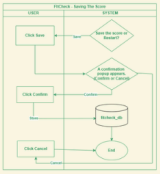


Fig. 4. Saving the score

IV. Experiments and Test Case Results

The FitCheck AI Gym Trainer System underwent a test case to validate its real-time posture classification and feedback mechanisms. The system was tested on various squat postures, using Mediapipe for pose detection and a pre-trained TensorFlow model for classification. The system was successful in classifying postures correctly, providing auditory cues, and maintaining an accurate count of improper postures. However, failure conditions like misclassifications, improper count errors, or system crashes were monitored to assess limitations.

Test Case	Expected Outcome	Actual Outcome	Pass/Fail
User performs a correct squat	System classifies as "Correct Form"	Correctly classified	Pass
User moves knees forward	System detects "Knee Ahead" error	Correctly classified	Pass
User rounds back incorrectly	System detects "Back Wrongly Positioned" error	Correctly classified	Pass
Audio feedback plays for errors and correct squat	Correct audio plays	Correct audio output for proper and improper squat	Pass

Table 1. Test Case Results

V. Client Feedback

After the walkthrough of the system, the user found FitCheck to be extremely useful, particularly for people working out alone who do not have a coach watching their form. The interface looked simple, and the real-time feedback was straightforward to follow; however, they were curious how intuitive it would be in practice. The usability and functionality of the feedback regarding squat posture were very clear, showing mistakes in leaning forward too much or not squatting deep enough. The real-time posture correction feature was the most appealing since it could be very useful for beginners who are uncertain about their form. FitCheck stood out as interactive, making it feel like a virtual coach. The user added that they would like audio feedback to avoid checking the screen too much, and a feature that tracks the progress of how things are getting better over time. They would also like the inclusion of a workout plan aimed at improving squat form. The user said that they would like to use it in public because they would give it a shot as long as it was still easy to set up and use. Their final suggestion was to extend the app beyond squats to more exercises, which would add so much value to the users.

VI. Conclusion

In conclusion, FitCheck has shown to be an effective system for improving exercise safety and posture correction through real-time computer vision feedback. Addressing the challenge posed by the client and meeting SDG 3: Good Health & Well-being, the system encourages safe workout habits and minimizes injury risks. The interactive features of FitCheck, according to user feedback, were good and users suggested audio feedback, tracking progress, and exercise options as recommendations for improvement, which further showed its potential for improvement and expansion.

References (Optional)

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Link: Group 11 - FitCheck_IEEE Poster.ppt

Analysis and Results

The FitCheck system was trained on 500 samples per class, using 50 epochs, a batch size of 16, and a learning rate of 0.001. Performance was evaluated using metrics such as accuracy, precision, and recall. The confusion matrix results indicated strong classification accuracy with minimal errors, demonstrating the system's effectiveness in posture classification.

- **Client Feedback**

Pros:
- Real-time posture correction provides immediate feedback.
- Clear visual feedback on squat form and improvements.
- Intuitive and user-friendly interface.

Cons:
- No audio feedback for hands-free guidance.
- Lacks a tracking feature to monitor progress over time.
- Currently limited to squat posture correction, without support for other exercises.

The client found FitCheck helpful, especially for individuals who work out alone and lack external guidance on squat form. The real-time posture correction feature was considered the most valuable, making the app feel like a virtual coach that ensures proper form.

- Usability & Functionality

- The interface was straightforward and easy to understand.
- The system effectively identified squat mistakes, such as leaning forward, incorrect knee alignment, and improper squat depth.
- The real-time corrections were particularly beneficial for beginners who struggle with proper form.

V. References *(including AI assisted tasks)*

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